Computer Science Department - University of Batna 2
2INF - Information Systems
Tutorial 1: The Enterprise and its Information System
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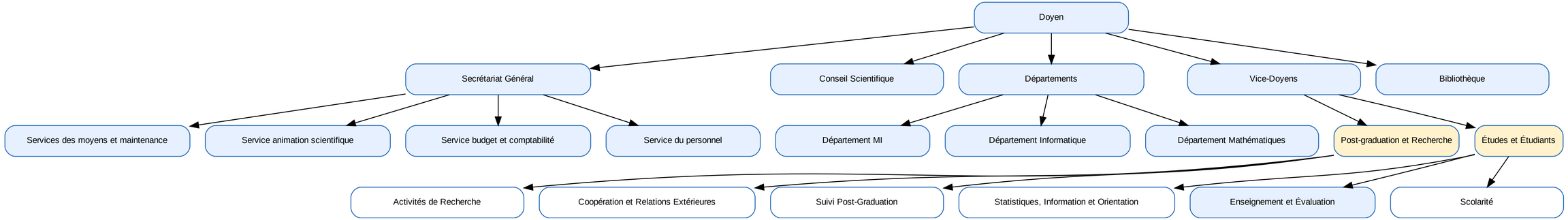
1 Exercise 1: Solutions

SOLUTION Question 1: System Identification

Identified Functional Systems

1. **Educational Management System**
   - **Mission**: Organize and deliver teaching
   - **Services**: Departments + Teaching and Evaluation Service
   - **Input**: Educational programs, enrolled students
   - **Output**: Delivered courses, assessments, diplomas
2. **Scientific Research System**
   - **Mission**: Develop research and innovation
   - **Services**: Vice-Dean Research + Research Activities Service
   - **Input**: Research projects, funding
   - **Output**: Publications, patents, collaborations
3. **Administrative Management System**
   - **Mission**: Provide administrative support
   - **Services**: General Secretariat (HR, Budget, Resources)
   - **Input**: Administrative requests, financial resources
   - **Output**: Administrative decisions, resource management
4. **Student Life System**
   - **Mission**: Animate university life
   - **Services**: Activities Service + Orientation Service
   - **Input**: Student needs, proposed activities
   - **Output**: Events, clubs, student welfare
5. **Documentation System**
   - **Mission**: Manage information resources
   - **Services**: Central Library
   - **Input**: Books, digital resources
   - **Output**: Loans, consultations, training

SOLUTION Question 2: Organizational Chart


Doyen
Secrétariat Général
Conseil Scientifique
Départements
Vice-Doyens
Bibliothèque
Services des moyens et maintenance
Service animation scientifique
Service budget et comptabilité
Service du personnel
Département MI
Département Informatique
Département Mathématiques
Post-graduation et Recherche
Études et Étudiants
Activités de Recherche
Coopération et Relations Extérieures
Suivi Post-Graduation
Statistiques, Information et Orientation
Enseignement et Évaluation
Scolarité

SOLUTION Question 3: Flow Analysis

Document Analysis

Document	Concerned System	Input/Output	Processing Service
Registration application	Educational Management	**Input**	Academic Affairs Service
Student certificate	Educational Management	**Output**	Academic Affairs Service
Student ID card	Educational Management	**Output**	Academic Affairs Service

Explanations:

- **Registration application**: Document provided by the student triggering the registration process
- **Student certificate**: Document produced after registration validation
- **Student ID card**: Material proof of belonging to the faculty

SOLUTION Question 4: Institution's Environment

External Actors and Relationships

Category	Actors	Nature of Relationships
Oversight authorities	Ministry of Higher Education, Rectorate	Regulatory framework, funding, diploma validation
Academic partners	Universities, Laboratories	Collaborative research, student exchanges, joint supervision
Socio-economic actors	Companies, Local authorities	Internships, employment, continuing education, partnerships
Users and beneficiaries	Students, Families	Education, orientation, student life, services
Suppliers	Publishers, IT companies	Educational resources, equipment, maintenance

Environment Analysis:

- **Regulatory environment**: Constraints and frameworks imposed by authorities
- **Academic environment**: Collaboration and knowledge sharing
- **Economic environment**: Professional opportunities and partnerships
- **Social environment**: Relations with the student community
- **Technological environment**: Solution and equipment providers

SOLUTION Question 5: IS, MS and OS Systems

Information

Information Systems (IS)	Operational Systems (OS)	Management Systems (MS)
SIS: Academic management, grades, schedules	**Teaching management**: Course scheduling, exam organization	**Executive dashboards**: Performance indicators
HRIS: Staff management, payroll, careers	**Infrastructure maintenance**: Classroom and equipment maintenance	**Scientific Council**: Research orientation, evaluation
Research IS: Projects, publications, laboratories	**Student activities**: Organization of cultural events	**Pedagogical committees**: Training improvement
Library IS: Library catalog, loans	**Library management**: Reception, loans, consultations	**Management control**: Budget monitoring, variance analysis

Relationship between systems:

$$\text{OS} \xrightarrow{\text{Raw data}} \text{IS} \xrightarrow{\text{Structured information}} \text{MS} \xrightarrow{\text{Strategic decisions}} \text{OS improvement}$$

Concrete example:

- **OS**: The academic affairs service records student grades (raw data)
- **IS**: The SIS calculates averages and generates statistics (information)
- **MS**: Management analyzes success rates to adjust programs (decisions)
- **Improvement loop**: New decisions improve operational functioning

2 Exercise 2: Solutions

SOLUTION Question 1: Process Analysis

Current System Process Analysis

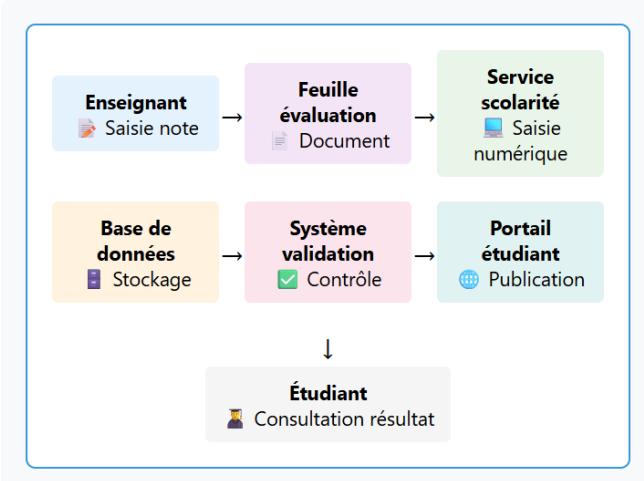
Process	Concerned Actors	Documents Used	Identified Problems
Grade entry	Teachers, Academic affairs service	Paper evaluation forms	Manual entry, error risks, long delays
Result validation	Program coordinators, Department council	Minutes, grade reports	Slow process, lack of traceability
Grade publication	Academic affairs service, Students	Notice boards, paper reports	Limited access, lack of confidentiality
Document archiving	Archives service, Secretariat	Physical files, binders	Clutter, search difficulties

Major problems of current system:

- **Double entry**: Error risk between paper and digital support
- **Significant delays**: Physical transmission time between services
- **Lack of transparency**: Unable to track grade status
- **High costs**: Printing, physical archiving, labor

SOLUTION Question 2: Flow Modeling

Flux d'information pour la gestion d'une note d'examen :



System Information Flow

Flow explanations:

1. Initial entry: Teacher assigns grade on paper support
2. Transmission: Evaluation form transmitted to academic affairs service
3. Digitization: Academic affairs service enters grade into system
4. Storage: Grade recorded in database
5. Validation: Coordinator validates grade batch
6. Publication: Grades made accessible to students
7. Consultation: Student consults result online

SOLUTION Question 3: New System Specifications

Detailed Functional Specifications

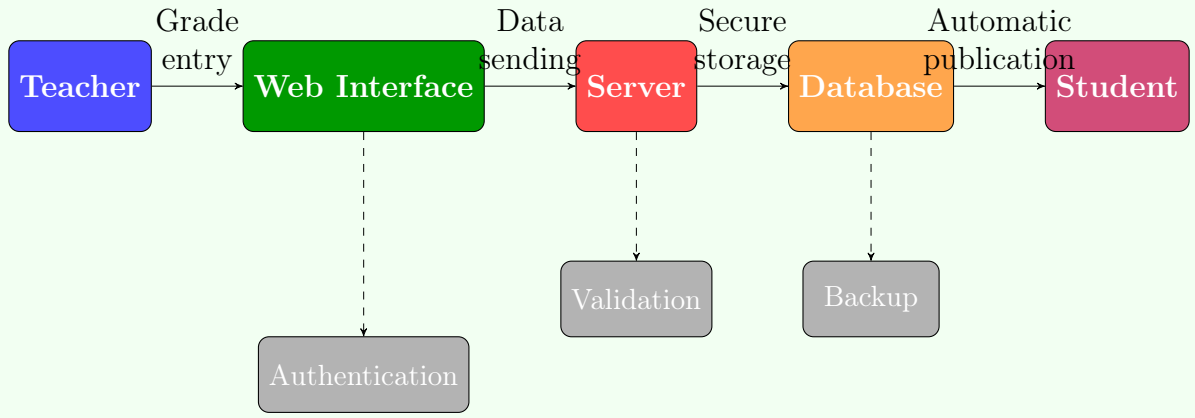
Actor	**Permissions**	**Constraints**
Teacher	Grade entry, statistics consultation, report editing	Limited period, assigned modules only
Program coordinator	Result validation, full consultation, reporting	Concerned program only, respect schedule
Academic affairs service	Full administration, official document editing	Respect procedures, hierarchical validation
Student	Personal grade consultation, transcript download	Personal data only, defined period

Technical characteristics:

- **Responsive interface**: Adapted desktop/tablet/mobile
- **Security**: Strong authentication with roles and permissions
- **Availability**: 24/7 access with automatic backups

SOLUTION Question 4: New System Information Flow

Optimized Grade Management System



Optimized flow description:

1. **Authentication**: Teacher accesses secure interface
2. **Direct entry**: Grade entry via web form with validation
3. **Server processing**: Automatic verification and average calculation
4. **Secure storage**: Recording in database with backup
5. **Immediate publication**: Instant access for students

Benefits:

- Delay reduced from several weeks to a few minutes
- Elimination of manual entry errors
- Complete process traceability
- 24/7 access from any connected device

SOLUTION Question 5: Advantages and Indicators

Advantages and Performance Indicators

For teachers	For students	For administration
Simplified and fast entry	Immediate access to results	Error reduction
Automatic calculations	Real-time information	Significant time saving
Intuitive interface	Confidentiality ensured	Complete traceability
Reduced administrative load	Online complaints	Cost reduction
Automatic statistics	Multi-device access	Better planning

Expected quantifiable benefits:

- Processing time reduction
- Error decrease
- Productivity gain
- User satisfaction
- Cost reduction